

Effect of Cultivar and Planting Pattern on Yield and Apparent Harvest Index in Soybean¹

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ABSTRACT

Three determinate soybean [*Glycine max* (L.) Merr.] cultivars were recently released because of their yield superiority to indeterminate cultivars when planted in narrow rows. The objective of our study was to determine if this yield advantage was because determinate cultivars had higher apparent harvest indices (ratio of seed yield to mature plant weight) than indeterminate cultivars when planted in narrow rows. Eight indeterminate and three determinate cultivars were grown in three environments with two planting patterns. The wide-row planting pattern had 76-cm rows with 38.9 viable seeds/m² and the narrow-row planting pattern had 19-cm rows with 52.7 viable seeds/m². Characters examined were maturity, mature plant population, seed yield, and apparent harvest index. Only seed yield had a significant cultivar by planting pattern interaction. Several indeterminate and the three determinate cultivars showed a high yield response to the narrow-row planting pattern. An 18% average yield increase resulted from decreasing row widths from 76 to 19 cm. The apparent harvest indices of the determinate cultivars were similar to the apparent harvest indices of several indeterminate cultivars in both planting patterns. Other factors such as total dry matter production and final plant population may be as important as apparent harvest index in determining seed yield for determinate cultivars.

Additional index words: *Glycine max* (L.) Merr, Determinate cultivars.

APPARENT harvest index in soybean [*Glycine max* (L.) Merr.] is defined as the ratio of seed yield to mature plant weight. Schapaugh and Wilcox (11) found a highly significant correlation ($r = 0.96$) between actual harvest index, which included leaves, stems, seeds and pods, and apparent harvest index. This indicated that apparent harvest index adequately reflected the actual harvest index of a genotype. Their study included genotypes of maturity Groups I to IV with indeterminate, semideterminate, and determinate genotypes planted in wide (76 cm) rows. They found no relationship between growth habit and the harvest index of a genotype. Mullen et al. (9) conducted a study comparing physiological characters of old vs. new cultivars. They found that the newer cultivars, Cumberland and Pella, had higher harvest indices than the older cultivars, Illini and Lincoln.

Three recently released determinate cultivars Gnome, Sprite, and Hobbit are generally superior in yield to indeterminate cultivars when planted in narrow (18 cm) row widths (5, 6, 7, 13). The objective of our study was to determine if this yield advantage was because determinate cultivars had higher appar-

ent harvest indices than indeterminate cultivars when planted in narrow rows. Seed yields and apparent harvest indices were determined for eight indeterminate and three determinate cultivars in two planting patterns.

MATERIALS AND METHODS

Eleven recently released soybean cultivars representing maturity Groups II and III were used in this study (Table 1). The cultivars were grown in three environments, Hoytville, Ohio in 1980, Wooster, Ohio in 1981, and Croton, Ohio in 1982. Planting and harvest dates were 7 May and 17 Oct., 1980, 18 May and 27 Oct., 1981, and 11 May and 13 Oct., 1982. Plots were either four rows 76 cm in width or eight rows 19 cm in width with both plots 6.1 m long. A 4.9-m portion of the two center 76-cm and four center 19-cm rows were harvested. The soybeans at each environment were planted in a split-plot arrangement of a randomized complete block with five replications. Whole plots were cultivars and subplots were the two planting patterns. All plots were planted with a unit planter and harvested with a plot combine. The seeding rate was 38.9 viable seeds/m² in 76-cm rows and 52.7 viable seeds/m² in 19-cm rows.

The following characteristics were measured on each plot: 1) maturity date, when 95% of the pods were their mature color; 2) plant population at maturity; 3) seed yield, oven-dried weight adjusted to 13.5% moisture; and 4) oven-dried stems and pod walls collected from the rear of the combine. In the combined analyses of variance, environments were considered random, and cultivar and planting patterns were considered fixed effects.

RESULTS AND DISCUSSION

No significant environment by cultivar by planting pattern interactions were found in the analysis of variance for any character. Only seed yield had a significant cultivar by planting pattern interaction ($P < 0.05$). In Maturity Group II, 'Vickery', 'Amcor', and Gnome responded more to the narrow-row planting pattern than did 'Beeson 80' and 'Century' (Table 1). In Maturity Group III, Hobbit, Sprite, and 'Asgrow A3127' responded more than Pella, Cumberland, and 'Williams 79'. Other researchers have found similar interactions (3, 4, 8). Costa et al., (8) in a Wisconsin study reported that 'Wells' responded less than 'Corsoy' to narrow (27-cm) rows. Cooper (3) in an Illinois study found that Corsoy responded more to 17-cm rows than 'Williams'. Cooper (4) later reported that Sprite also responded more to 17-cm rows than Williams.

A highly significant 18% yield difference ($P < 0.01$) occurred between the wide- and narrow-row planting pattern. Yields were consistently higher for the narrow-row planting pattern over the three environments with a 20% response at Hoytville in 1980, a 15% response at Wooster in 1981, and an 18% response at Croton in 1982. This agrees with data from a number of other recent studies (3, 4, 8, 12). Costa

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Table 1. Cultivar by planting pattern means for maturity, mature plant population, seed yields, and apparent harvest index.

Cultivar plant- ing pattern†	Growth habit†	Maturity		Mature plant population		Seed yield		Apparent harvest index‡	
		W	N	W	N	W	N	W	N
		Sept. date		plant/m²		—Mg/ha—		—%—	
Maturity Group II									
Vickery	I	13	13	26.7	36.7	2.96	3.77	57.8	56.6
Beeson 80	I	16	16	25.6	35.7	3.05	3.45	56.9	53.0
Century	I	17	17	29.2	36.0	3.28	3.68	57.3	54.6
Amcor	I	19	19	26.1	33.9	3.20	3.78	56.8	55.9
Gnome	D	19	19	28.1	36.6	3.06	3.87	55.9	54.7
Maturity Group III									
Pella	I	20	20	28.9	39.3	3.15	3.62	56.6	54.1
Hobbit	D	22	22	29.2	39.5	3.29	3.94	58.3	55.1
Sprite	D	22	22	30.1	40.5	3.26	3.96	58.4	55.7
Asgrow A3127	I	23	23	28.1	38.5	3.36	4.01	57.4	55.4
Cumberland	I	24	24	28.6	34.7	3.27	3.72	58.4	55.9
Williams 79	I	24	24	26.0	35.3	3.05	3.47	55.5	52.9
LSD (0.05)¶		1		2.8		0.22		1.5	

† I = indeterminate and D = determinate.

‡ W = 76-cm rows seeded at 38.9 viable seeds/m², and N = 19-cm rows seeded at 52.7 viable seeds/m².

§ (Ratio of seed yield to mature plant weight) × 100.

¶ To compare cultivar means within a planting pattern and planting pattern means within a cultivar.

et al. (8) reported a consistent yield advantage which averaged 21% for 27-cm rows as compared to 76-cm rows in a 3 year Wisconsin study. Cooper (3) found a 20% yield response from 17-cm rows as compared to 75-cm rows over five Illinois environments. Cooper (4) also reported an 18% yield response to narrow rows for 'Beeson', Williams, and 'Elf' in two dry Illinois environments and a 16% response in two irrigated environments. Taylor (12) obtained a 17% greater soybean yield from 25-cm rows vs. 100-cm rows in a year of above normal precipitation in Iowa, but found no row spacing effects in drier years.

A highly significant difference ($P < 0.01$) between the wide- and narrow-row planting pattern also occurred for apparent harvest index (Table 1). However, in contrast to seed yield, apparent harvest indices were higher in the wide rows than in the narrow rows. This may be due to the different plant populations. The mature-plant populations were significantly lower ($P < 0.01$) in the wide rows than in the narrow rows, which were related to the different seeding rates. Several researchers (10, 14, 15) have reported that lower plant populations will increase harvest index.

The highest-yielding cultivars in the narrow-row planting pattern were Asgrow A3127, Sprite, and Hobbit, while the lowest-yielding cultivars were Beeson 80 and Williams 79. The same cultivars also were among the highest and lowest for apparent harvest index. However, Vickery, Amcor, and Cumberland, which had the highest apparent harvest indices in the narrow-row planting pattern, were not the highest-yielding cultivars. It appears that a high yielding cultivar will have an above average apparent harvest index (9), but a cultivar with a high apparent harvest index will not always have high yield.

Vickery and Williams 79 had the highest and lowest apparent harvest indices, which agrees with data reported by Beaver and Cooper (1) and Schapaugh and Wilcox (11) on the cultivars 'Corsoy' and Williams. Vickery and Williams 79 were derived from Corsoy and Williams through backcrossing, thus, they would be expected to perform similarly to their recurrent parents.

Buzzell and Buttery (2) and Schapaugh and Wilcox (11) found high apparent harvest indices to be strongly associated with early maturity. This relationship was not observed when Maturity Group II cultivars were compared to Maturity Group III cultivars in this study.

In summary, the three recently released determinate cultivars, Gnome, Sprite, and Hobbit were among the highest-yielding cultivars in narrow rows. The apparent harvest indices for the determinate cultivars were similar to those of several indeterminate cultivars in narrow rows. Other factors such as total dry matter production and final plant population may be as important as apparent harvest index in determining seed yield for determinate cultivars.

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